MIL-M-9397E(USAF)
2 JULY 1979
SUPERSEDING
MIL-M-9397D(USAF)
28 May 1968

MILITARY SPECIFICATION

FREQUENCY CONVERTER, MOBILE, TYPE MC-1A

This specification is approved for use by AFLC CASO/LODS, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of mobile electrical power converter, designated Type MC-1A, which converts 60 Hertz power to 400 Hertz and direct current power.

2. APPLICABLE DOCUMENTS

2.1 Issue of document. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower 500 HP and Smaller.
QQ-P-416	- Plating, Cadmium (Electrodeposited).

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods Of.
DOD-D-1000	- Drawing, Engineering and Associated Lists.
MIL-W-5088	- Wiring, Aerospace Vehicle.
MIL-C-5541	- Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys.
MIL-C-5756	- Cable and Wire, Power, Electric, Portable.
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series, General Specification For.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6125

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

MIL-M-8090	- Mobility, Towed Aerospace Ground Equipment, General Requirements For.
MIL-A-8421	- Air Transportability Requirements, General Specification For.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-M-10304	- Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification For.
MIL-I-45208	- Inspection System Requirements.
MIL-C-83286	- Coating, Urethane, Aliphatic Isocyanate for Aerospace Applications.

STANDARDS

Military

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-143	- Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-195	- Marking of Connections for Electric Assemblies.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-701	- Lists of Standard Semiconductor Devices.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests; Exponential Distribution.
MIL-STD-808	- Finishes, Protective and Codes, for Finishing Schemes for Ground and Ground Support Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-831	- Test Reports, Preparation of.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1562	- Lists of Standard Microcircuits.
MS16658	- Connector, Plug and Duplex Receptacle, Electrical, Power, 2 Pole 3 Wire, Ground Type, 15 Ampere, 125 Volts.
MS20659	- Terminal, Lug, Crimp Style, Copper, Uninsulated, Ring Tongue, Type I, Class I.
MS90328	- Cable Assembly, External Electrical Power, Aircraft, 115/200 Volts, 400 Hertz.
MS90347	- Cable Assembly, External Electrical Power, Aircraft, 28V DC Operating Power.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

AIR FORCE-NAVY AERONAUTICAL

AN2552 - Receptacle - External Power, 28 Volt, DC.
AN3114 - Receptacle, External Power, 115/200 Volts, 3 Phase.

DRAWING

Air Force

7545[illegible]52 - Requirements for Finishes, Protective, and Codes for San Antonio ALC Ground, and Ground Support Equipment.

HANDBOOKS

Military

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposal shall apply.

ANSI S1.4-1971 (R1976) - Sound Level Meters.
ANSI S1.11-1966 (R1976) - Octave, Half Octave and Third Octave Band Filter Sets, Specification for.

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

USA STD Y32.2-1967 (R1975) - Graphic Symbols for Electrical and Electronics Diagrams.

(Application for copies should be addressed to United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for Preproduction Testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

3.2 Components. The frequency converter shall include the following:

Description	Requirement
Synchronous Motor	3.7
Alternating Current Generator	3.8
Solid State Direct Current Generator	3.9
Control Panel	3.10.2
Auxiliary Electrical Equipment	3.11

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as specified in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts and under the following conditions:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawing.

3.3.2 Standard parts. With the exception specified in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified by their part numbers.

3.3.2.1 Semiconductor devices shall be selected in accordance with MIL-STD-701. Microcircuits shall be selected in accordance with MIL-STD-1562. Electron tubes shall not be used.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service.

3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Rubber. Where rubber or similar material is used, it shall be compounded with a reaction product to improve its oil resistance, electrical insulating properties, and ability to withstand extreme low temperatures.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

3.5 Design and construction. The motor-generator shall be of brushless design and shall be constructed so that no parts will work loose in service. The converter unit shall be built to withstand the strains, shocks, vibrations and other conditions incident to shipping, storage, installation, and service.

3.5.1 Safety. The converter shall be so designed as to positively prevent accidental contact by personnel with rotating parts or potentially dangerous electrical voltages. Warning legends shall be used where appropriate. Overload and overheat sensing devices shall be included that will prevent destruction of the equipment when conditions beyond the design range occur. The unit shall comply with the provisions of MIL-STD-454, Requirement 1.

3.5.2 Reliability. The frequency converter set shall have an upper test level mean time between failure (MTBF), as defined in MIL-STD-781, of 4000 hours and a lower test level MTBF of 1000 hours. In addition, no depot level repair shall be required prior to 8000 hours of operation. For reliability purposes, a failure shall be defined as any condition that causes the equipment to operate outside specification requirements defined by paragraphs 3.6 thru 3.9 of this document.

3.5.2.1 Parts derating. For electrical and electronic parts used in the equipment design, a derating policy at least as stringent as that specified in the Appendix to this specification shall be applied.

3.5.3 Maintainability. The frequency converter shall be designed for ease of repair at all levels of maintenance. Preventive maintenance shall be required no often than once each 400 operating hours. It shall require a maximum of 30 minutes clock time and one man hour of effort to complete.

3.5.3.1 Special tools and test equipment. No special tools or test equipment shall be required at organizational or intermediate level to repair this equipment. Depot level special tools and test equipment shall be permitted only if strong justification is presented explaining why items already in the Air Force inventory cannot be used.

3.5.3.2 Instruction books. Final or approved preliminary Technical Orders will be delivered concurrently with and accompany production level equipment. Government will not accept delivery of any production equipment if it is not accompanied by this technical data.

3.5.4 Electrical circuitry. The converter output circuits shall be as shown on Figure 1. Wire size shall be at least as large as the corresponding conductors in the output cable assemblies. All conductors shall be legibly and permanently marked in accordance with the cable identifying code in MIL-W-5088. Terminal blocks shall be appropriately marked to facilitate connection of the cable assemblies.

3.5.5 Component balance. The rotating components of the motor-generator shall be statically and dynamically balanced. Balancing shall be accomplished by drilling out material, by using noncorroding balancing weights

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

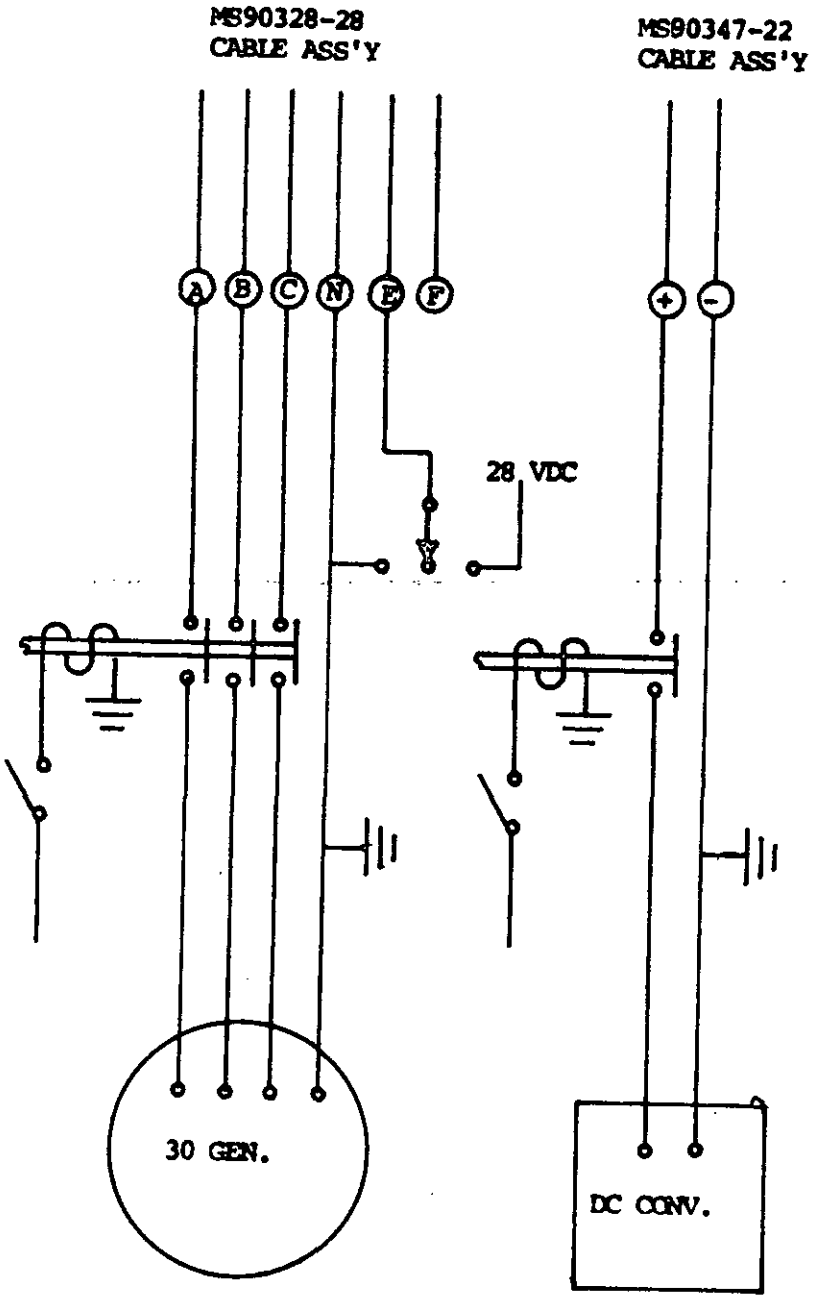


FIGURE 1. Output Circuits

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

that are positively locked, by riveting, by using lead or babbitt metal carried in a receiver in such manner as to preclude the possibility of its breaking loose, or by using steel weights securely welded in place. In no case shall the use of lockwashers be considered a positive means of locking balancing weights. The double amplitude of any linear vibration of the motor-generator shall not exceed 0.001 inch. All other components shall be secured as necessary to prevent them from vibrating.

3.5.6 Weatherproofing. The converter unit shall be of splash-proof construction as defined in MIL-STD-108. This requirement may be applied to the unit as a whole (see 3.2), or to the motor, alternating current generator, solid state direct current generator, and controls individually.

3.5.7 Mobility. The motor-generator shall meet the type I, group B, class 2 mobility requirements of MIL-M-8090 except that the maximum towing speed shall be 10 miles per hour.

3.5.7.1 Steering. Swivel-type steering shall be incorporated that will permit a maximum angle of 90 degrees.

3.5.7.3 Parking brake. The parking brake shall be capable of holding the motor-generator against a 15 degree slope. The brake shall be operated by lifting the towbar toward the vertical position. Brake action shall commence when the center of the lunette eye is between 40 and 48 inches above ground and shall be continued as the towbar is raised until the maximum braking action is obtained when the towbar is in vertical position. Means shall be provided to keep the brake engaged for parking purposes when the towbar is in the vertical position.

3.5.8 Transportability. The converter set shall meet the air transportability requirements specified in MIL-A-8421.

3.5.9 Electromagnetic interference. The converter set shall meet the electromagnetic interference requirements of MIL-STD-461 described in paragraph 4.6.6.

3.5.10 Locking devices. Wherever possible, screws, bolts, pins, et cetera, shall be furnished with Air Force standard locking devices. Safety wire, self-locking nuts, cotter pins, lockwashers, et cetera, will be acceptable, except for securing rotor balancing weights.

3.5.11 Bearings. All bearings supporting the rotating parts shall be sealed, lifetime-lubricated ball or roller bearings.

3.5.12 Lifting eyes. Suitable lifting eyes shall be provided on the motor and alternator to lift the components when being removed for maintenance.

3.5.13 Motor leads. The motor leads shall be brought out to a terminal board, arranged in sequence, and marked L_1, L_2, L_3 and L_0 in accordance with MIL-STD-195.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

3.6 Performance.

3.6.1 Ratings. The frequency converter set shall have the following simultaneous continuous output ratings:

a. 37.5 kilovolt-amperes, 0.8 power factor, 400 Hertz, 120/208 volts, 3 phase, 4 wire or 12.5 kva, 0.8pf, 400 Hz, 120 volts, single phase.

b. 500 amperes, 28 volts, direct current.

3.6.1.1 Intermittent capacity. The converter shall have continuous output ratings as specified plus an intermittent capacity of 125 percent of full load ratings.

3.6.1.1.1 Intermittent duty. The intermittent duty shall consist of 15 minute cycles of operation, 3 minutes of which shall be at 125 percent of full loads and 12 minutes at no loads.

3.6.2 Efficiency. The efficiency shall be at least 70 percent when the converter is operated at the ratings specified in 3.6.1 with rated input voltage in an ambient temperature of 50 degrees Celsius. Input and output cable losses shall be included in the efficiency calculations.

3.6.3 Environmental conditions. The converter shall be capable of withstanding the following conditions without detrimental effect to subsequent operation:

a. Ambient operating temperatures ranging from -40 deg C to +50 deg C and storage temperatures ranging from -55 deg C to + 55 deg C.

b. Heavy rainfall at an angle of 15 degrees to the vertical axis.

c. Fungus growth as encountered in tropical climates.

d. Sand and dust particles as encountered in desert areas.

e. Relative humidity up to and including 100 percent.

f. Exposure to salt fog.

3.6.4 Temperature rise. All windings of the converter shall operate as specified herein with a temperature rise not exceeding the following based on a 50 deg C ambient.

Insulation Class	Thermocouple Method	Resistance Method
A	55 deg C	65 deg C
B	75 deg C	90 deg C
F	90 deg C	117 deg C
H	105 deg C	140 deg C

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

3.6.5 Dielectric strength. Windings of 50 volts or less shall be capable of withstanding 500 volts root mean square (rms) for 1 minute or 600 volts rms for 1 second. Windings over 50 volts shall be capable of withstanding twice rated voltage plus 1,000 volts rms for 1 minute or 120 percent of the minute value for 1 second.

3.6.6 Noise levels. The noise levels produced by the converter shall not exceed the sound pressure level values specified in Table I with the equipment operating at design capacity.

TABLE I. Noise Levels

Octave Band Center Frequencies (Hz)	Sound Pressure Levels db ref. (0.0002 dynes/cm^2)
63	78
125	72
250	67
500	64
1000	62
2000	60
4000	58
8000	57

3.7 Motor. The motor shall be a 60 Hz, three phase, brushless, synchronous motor conforming to CC-M-1807. Coil leads of the motor shall be brought out to a terminal block for easy reconnection for either input voltage. The motor shall be of sufficient capacity to enable the motor-generator to deliver the rated output plus overload as specified herein. The temperature rise shall not be exceeded when the motor is supplied with rated voltage ±10 percent.

3.7.1 Controls. The motor shall be controlled by a single start and a single stop button. The circuitry shall be generally as shown on Figure 2 and shall enable the motor-generator to operate from the dead-cable system as shown on Figure 3. Starting circuitry shall also permit starting on a hot line. The controls shall include a starting contactor, field application relay, field discharge resistor, protective devices, and any other items necessary for proper operation. In all cases where start and stop or on and off push buttons are used, the stop or off button shall be supervisory.

3.7.1.1 Input contactor. The input contactor shall have positive action when the motor is started on input voltage as low as 80 percent rated voltage. The contactor shall not drop out nor chatter during a voltage drop due to starting that lowers the line voltage to 60 percent of rated value. If line voltage falls to less than 60 percent of rated value, the input contactor shall open without chatter and shall remain open.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

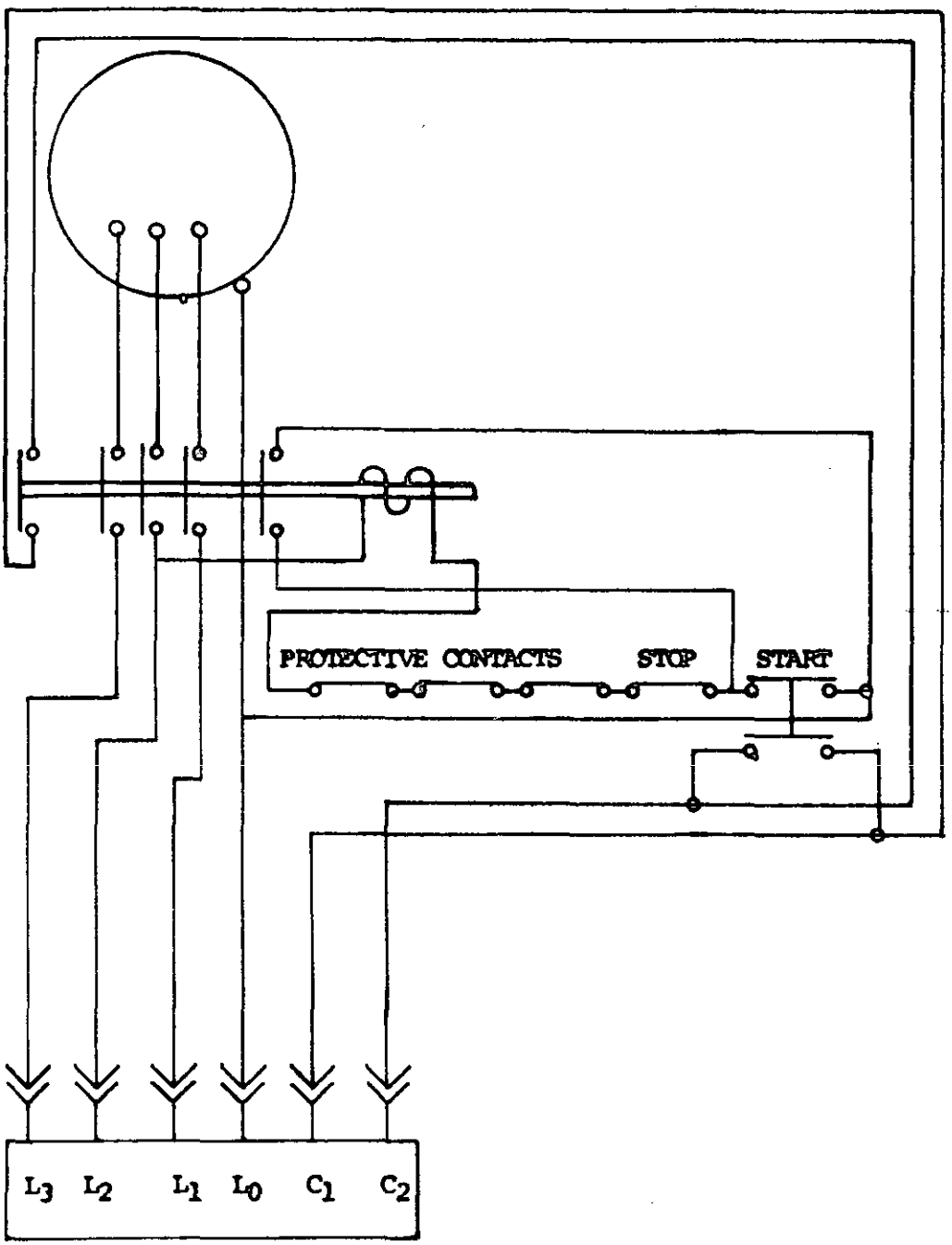


FIGURE 2. Schematic, Starting Control

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

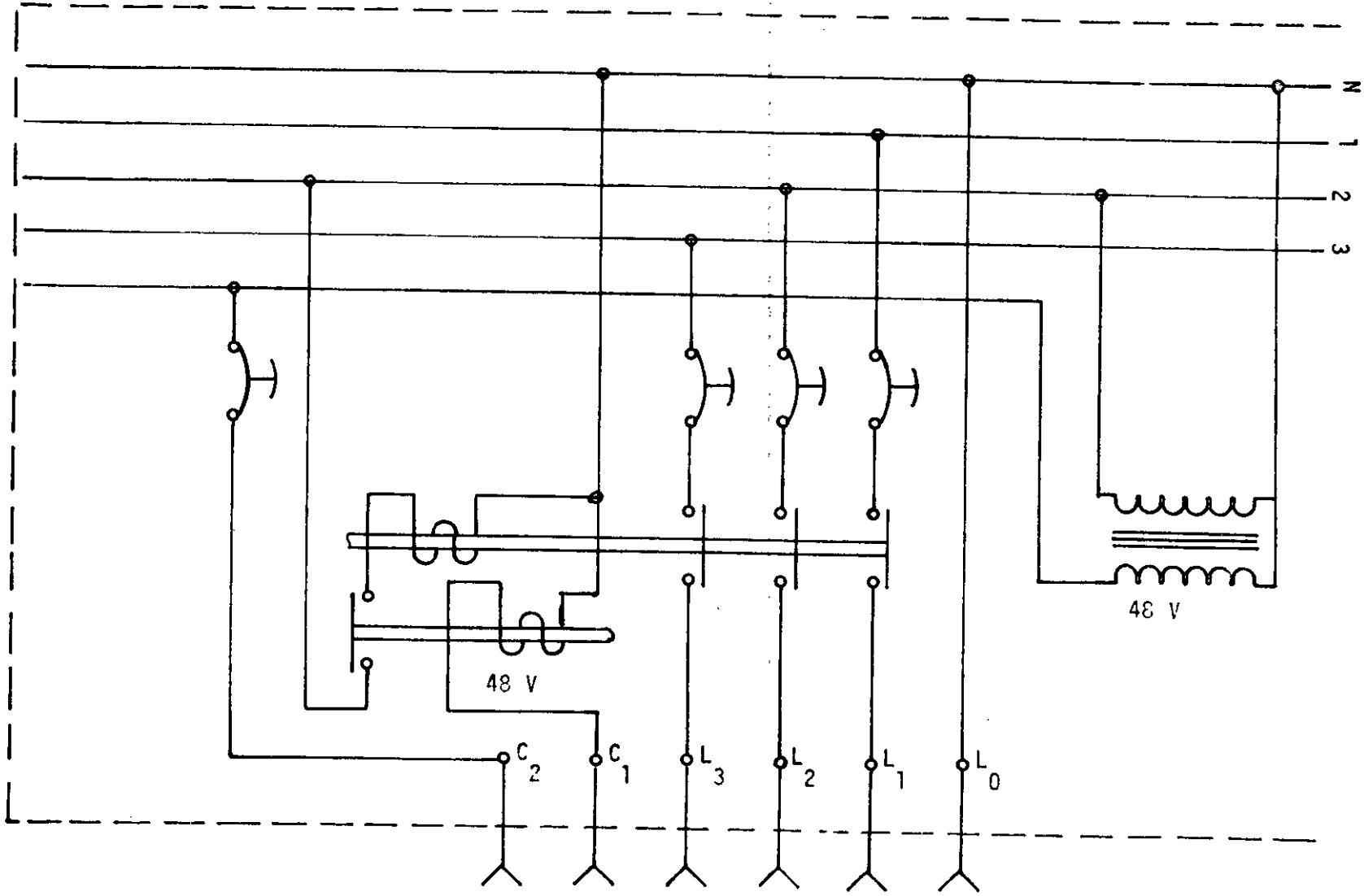


FIGURE 3. Input Power Panel Schematic

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

3.7.2 Protective devices. The following protective devices shall be provided:

a. Overload device to open the input contactor in event the input current becomes excessive. This device shall have an inverse time characteristic to prevent tripping on starting current, unless the starting period is unduly long.

b. Undervoltage device to open the input contactor in the event that the input voltage drops below 60 percent of rated input value.

3.7.3 Motor inrush current. The line current to the motor shall not exceed 275 amperes at 440 volts or 550 amperes at 220 volts after 0.05 second. Resistors shall not be used as a means of limiting inrush current.

3.8 Alternating current generator. The alternating current generator shall be a 37.5 kva, 0.8 pf, 120/208 volt, 400 Hz brushless alternator. It shall be built together with the motor as a two-bearing unit. It shall be wired in a wye connection with terminals designated T_1, T_2, T_3, and T_0. The phase sequence shall be in the order of T_1, T_2, T_3. The direction of rotation shall be clearly marked by an arrow. Overload capacity shall be as specified herein.

3.8.1 Frequency stability. The frequency of the output voltage shall be steady. The motor-generator shall have no tendency to hunt. Oscillation in the output frequency shall be damped out within 3 seconds after sudden application or removal of any combination of ac and dc loads including full rated ac and dc loads with the ac load at any rated power factor. This requirement shall also apply during sudden changes of input voltage including +10 percent of rated voltage while any combination of ac and dc loads at any rated power factor is being steadily supplied by the generators. Oscillations shall be considered to be damped out when the output frequency deviation diminishes to and remains less than $\pm\frac{1}{2}$ Hz with a period of oscillation longer than 1 second. Measurements of frequency shall be made by means of a frequency discriminator with the output recorded on an oscillograph, or by other means approved by the procuring activity.

3.8.2 Voltage regulator. A solid state voltage regulator shall be provided and shall maintain rated voltage ±1.00 percent throughout the load and power factor range. The voltage regulator shall incorporate a screwdriver adjustment for manually adjusting the regulated voltage between 90 and 110 percent of rated voltage in steps of not more than 0.5 volt. The adjustment mechanism shall have a locking device for preventing the voltage regulator setting from being changed accidentally. When the voltage has been set at rated value at no load, the voltage that is maintained by the regulator within the limits specified herein shall be the steady-state voltage for the load. Electron tubes shall not be used. The regulator shall meet the following criteria:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

a. The regulator shall consist of plug-in printed circuit board assemblies which are readily removable for repair. The printed circuit board plug-in connectors shall be keyed to prevent any mismating of boards. A universal extender board shall be provided with each set to facilitate trouble-shooting of the voltage regulator printed circuit boards while the set is operating. The circuits shall be contained in a ventilated housing provided with a removable cover and a single MS type connector containing both the input and output circuits to permit removal of the voltage regulator without disturbing the interface wiring.

b. The voltage regulator shall contain the circuitry to provide overvoltage and undervoltage protection for the connected load.

c. The voltage regulator shall contain an ajustable line drop compensation system.

d. The voltage regulator shall have an inherent response time of less than .001 second.

3.8.3 Harmonic content. The wave shape of the output voltage shall be essentially sinusoidal. The total harmonic content shall not exceed 2 percent for balanced loads throughout the load and power factor range. Total harmonic content in percent shall be one hundred times the square root of the sum of the squares of the effective values of all the harmonics contained in the wave divided by the effective value of the fundamental. Harmonic content shall be measured from line to line and from line to neutral.

3.8.4 Transient recovery. When full load is suddenly applied or removed, the voltage shall not vary from rated value by more than 10 percent. The voltage shall recover to and remain within 1 percent of the steady-state voltage within 0.2 second.

3.8.5 Voltage modulation. Voltage modulation shall not exceed 2 percent. Voltage modulation in percent shall be calculated by the following formula:

$$\frac{100\ (E\ max - E\ min)}{E\ max + E\ min}$$

3.8.6 Phase balance. With the generator wye-connected, and balanced three phase 0.8 power factor full load applied, the voltage unbalance shall not exceed 1 percent. The voltage unbalance at no load shall not exceed 1 percent. With 0.8 power factor full load applied to any phase and no load on the other phases, the voltage unbalance shall not exceed 3 percent. The voltage unbalance shall be calculated by comparing the maximum deviation of any phase-to-neutral voltage from the average phase-to-neutral voltage with the average phase-to-neutral voltage.

3.8.7 Power factor. The motor-generator shall provide required voltage regulation, harmonic content and all other characteristics specified herein while supplying 30KW load from 0.8 lagging pf to 1.0 pf.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

3.8.8 Contactor. The alternator shall be connected to the output cables through an electrically operated contactor essentially as shown on Figure 1. Controls for the contactor shall be mounted on the control panel.

3.8.8.1 Output. The output contactor shall disconnect the generator from the output cables. The contactor may incorporate a protective device or may be tripped by the operation of one or more of the protective devices specified herein.

3.8.9 Alternating current protective devices. The output contactor shall be electrically interlocked with the input contactor so that if the input contactor opens, the output contactor will open and remain open. The output contactor control circuitry shall be such that the output contactor cannot be closed unless the motor has synchronized and output voltage has been established. The following protective devices shall be provided:

a. Overload device to disconnect the generator from the output cables if the load current exceeds 130% of the full load current except during transient conditions.

b. Overvoltage device to disconnect the generator from the output cable if output voltage exceeds 110% rated voltage at the terminals. Provisions shall be made for a time delay to prevent tripping during the specified allowable transient conditions.

c. Undervoltage device to disconnect the generator from the output cables if the output voltage goes below 90% rated voltage at the terminals. Provisions shall be made for a time delay to prevent tripping during the specified allowable transient condition.

3.9 Solid state direct current generator. A solid state direct current generator shall be included as part of the unit. This generator shall supply 500 amperes at 28 volts direct current and shall have the overload capacity specified herein.

3.9.1 Voltage regulator. A solid state voltage regulator shall be provided and shall maintain rated voltage within 2% throughout the load range. The voltage regulator shall incorporate a screwdriver adjustment for adjusting the output voltage from 26 to 30 volts in steps of not more than 0.25 volt. The adjustment mechanism shall have a locking device for preventing the voltage-regulator setting from being changed accidentally. After rated voltage has been set at no load, the voltage that is maintained by the regulator at any other load specified herein shall be the steady voltage for that load. Electron tubes shall not be used. The regulator shall meet the following criteria:

a. The regulator shall consist of plug-in printed circuit board assemblies which are readily removable for repair. The printed circuit board plug-in connectors shall be keyed to prevent

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

any mismating of boards. A universal extender board shall be provided with each set to facilitate trouble-shooting of the voltage regulator printed circuit boards while the set is operating. The circuits shall be contained ina ventilated housing provided with a removable cover and a single MS type connector containing both the input and output circuits to permit removal of the voltage regulator without disturbing the interface wiring.

b. The voltage regulator shall contain the circuitry to provide overvoltage and undervoltage protection for the connected load.

c. The voltage regulator shall contain an adjustable line drop compensation system.

d. The voltage regulator shall have an inherent response time of less than .001 second.

3.9.2 Ripple. The direct current ripple voltage shall not exceed a peak to peak value of 1 volt throughout the load range.

3.9.3 Transient response. When full load is suddenly applied or removed, the output voltage shall not vary from the steady-state voltage by more than 10 percent. The voltage shall recover to and remain within 2 percent of the steady-state value within 0.2 second.

3.9.4 Output contactor. An electrically operated output contactor shall disconnect the generator from the output cables. This contactor may incorporate a protective device or may be tripped by the operation of one or more of the protective devices specified herein. Controls for the contactor shall be mounted on the control panel.

3.9.5 Direct current protective devices. The output contactor shall be electrically inter-locked with the input contactor so that if the input contactor opens, the output contactor will open and remain open. The following protective devices shall be provided:

a. The generator shall be self-protected from excessive current demand.

b. An overvoltage device to disconnect the generator from the output cables if the output voltage exceeds 32 volts at the terminals longer than the specified recovery time.

c. An undervoltage protective device to disconnect the generator from the output cables if the output voltage goes below 24 volts at the terminals longer than the specified recovery time.

3.10 Control cabinet. A control cabinet or enclosure shall be provided in which the control panel and all contactors, relays, regulators, controls, and other components necessary for control and operation of the converter set shall be mounted. The control cabinet shall be held to minimum size

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

consistent with accessibility and cooling of components. The sides shall be hinged or removable to permit easy access to components therein.

3.10.1 Control panel. The control panel shall contain all meters, controls and test jacks. The panel shall be hinged to permit easy access to the space behind it. All items on the control panel shall be concisely and legibly labeled and shall be grouped in a logical and orderly fashion.

3.10.1.1 Meters. The following instruments, a through e, shall conform to the applicable requirements of MIL-M-10304.

a. Voltmeter MR36W500ACVVR for 60 Hz input, connected on the power side of the input contactor.

b. Voltmeter MR36W150AFVVR for 400 Hz output.

c. Voltmeter MR36W030DCVVR for direct current output.

d. Ammeter MR36W150AFAAR for 400 Hz output.

e. Ammeter MR36W800DCAAR for direct current output.

f. Frequency meter (scale not to exceed 390 to 410 Hz nor be less than 395 to 405 Hz.

g. Hourmeter (scale range 0 to 99,999.9 hours) connected on the motor side of the input contactor.

3.10.1.1.1 Mounting. All meters shall be mounted in vibration-resistant mounting to protect them from vibration incident to normal operation and towing.

3.10.1.2 Transformers and shunts. Current transformers and shunts shall be provided where necessary for proper functioning of the meters.

3.10.1.3 Relay control switch. The motor starting switch shall contain contacts to energize the safety (dead cable) relay in the input power panel, as indicated in Figures 2 and 3.

3.10.1.4 Selector switches. The alternating current output voltmeter and ammeter shall be provided with selector switches so that any of the three phase-to-neutral voltages or phase currents can be read.

3.10.1.5 Test jacks. The test jacks shall be provided immediately below the ac and dc output voltmeters for checking the output voltages with portable voltmeters. Jacks shall be designed to mate with pointed test probes.

3.10.2 Instruction manual and record compartment. The control cabinet shall be provided with a compartment for the storage of instruction manuals and equipment maintenance and inspection records. If not contained inside the cabinet, the compartment shall be waterproof and shall have minimum dimensions of 9 x 11½ x 2 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

3.11 Auxiliary electrical equipment.

3.11.1 Receptacles. All receptacles shall be clearly marked with electrical output characteristics. The receptacles shall have spring-loaded, weatherproof covers to prevent accidental contact with the electrical contacts and to protect them from weather. The receptacles shall be so mounted under a rain shield as to be easily accessible, without opening a door, and protected from driving rain.

3.11.1.1 Standard receptacles. Receptables conforming to MS16658 and having ratings as specified shall be provided:

Ratings	Quantity
120 volt, 60 Hz, 15 amp, single phase	1
120 volt, 400Hz, 15 amp, single phase	1
28 volt, dc, 15 amp	1

3.11.1.2 A receptacle conforming to AN3114 shall be provided and connected in parallel with the alternating current generator output cable.

3.11.1.3 A receptacle conforming to AN2552 shall be provided and connected in parallel with the direct current generator output cable.

3.11.2 Wiring. All wiring except output wiring shall be of the extra flexible type and of sufficient size to carry the required current. Wires shall be marked similar to the marking requirements of MIL-W-5088. Each wire should be identified over its entire length at intervals not exceeding 15 inches and within 3 inches of its terminating point or junction (except permanent splices). The identification shall be permanently imprinted or banded on the wire so that ties, clamps or coverings will not have to be removed to read the identification. Symbols shall be in accordance with USA STD Y32.2.

3.11.2.1 Securing wires and parts. Current-carrying wires and parts shall be secured to prevent rubbing contact with each other and with other parts of the converter set.

3.11.3 Cables. Cables shall be provided and attached to the converter set in such manner that they will not interfere with personnel attending the control panel.

3.11.3.1 Input cable. Power shall be delivered to the frequency converter set by means of a six conductor cable assembly consisting of MIL-C-5756 wires, MIL-C-38159 male plug connector (MS27265) at one end, MIL-T-7928 lug terminals (MS20659) at the other end, non-metallic banding material and non-metallic terminal identification sleeving. The cable shall be 75 feet long. 0 gauge conductors shall connect to the L_1, L_2, and L_3 contacts of the connector, 4 gauge conductor shall connect to the L_0 contact, and 12 gauge connectors shall connect to the C_1 and C_2 contacts. The six wires shall be banded together at approximately two foot intervals throughout the cable length. Terminal identification sleeves shall be applied to the six wires within 6 inches of the lug terminals. The cable shall be attached to the frame of the frequency converter set to prevent strain on the terminal board.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

3.11.3.2 Output cables. Outputs of the converter set shall be delivered to the aircraft by means of the following cable assemblies:

a. MS90347-22 2 Conductor Cable

b. MS90328-28 6 Conductor Cable

3.11.3.3 Connections. Cables shall be connected to the converter set through terminal strips as shown on Figure 1. Provisions shall be made for anchoring one each additional dc and ac output cable.

3.11.3.4 Cable stowage. Provisions shall be made for stowing the input and output cables without disconnecting them from their respective terminals.

3.11.4 Terminals. Input and output stud-type terminals which will accommodate lug terminals conforming to MS20659 shall be provided on the converter set terminals blocks. Input, ac output, and dc positive output terminals shall have 3/8-24 unified fine thread (unf) 2A screw threads. Direct current negative output terminals shall have ½-20 unf-2A screw threads. All terminals shall be marked as specified herein and so grouped as to be easily identified. A cover shall be provided to prevent actual contact with the studs when input and output cables are connected to the converter. Two nuts with washers shall be provided on each terminal. Output terminal studs shall be of sufficient length to accomodate two sets of cables.

3.12 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

3.13 Dimensions. The dimensions of the frequency converter set shall not exceed the following:

a. Length - 108 inches
b. Width - 58 inches
c. Height - 40 inches

3.14 Weight. The weight of the complete unit shall not exceed 4,500 pounds.

3.15 Finishes and protective coatings. Cleaning, painting, plating, anodic films and chemical treatments shall be in accordance with MIL-STD-808 as supplemented by Air Force Drawing 7545352 (San Antonio Air Logistics Center). Cadmium plating in accordance with QQ-P-416 shall be used where applicable only when no other protection process can reasonably provide the required protection. The use of cadmium plating shall require approval of the procuring activity.

3.16 Marking.

3.16.1 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:19Z
Check the source to verify that this is the current version before use.

special marking shall be added:

AC Input: 220/440V, 60 Hz, 3 phase.
AC Output: 120/208V, 400 Hz, 3 phase, 4 wire, 104 amp.
DC Output: 28V, 500 amp.

3.16.2 Plaques and lettering. Plaques and lettering shall be provided as specified in the following paragraphs.

3.16.2.1 Towing speed. The lettering "MAX TOWING SPEED 10 MPH" shall be stenciled near the tow bar.

3.16.2.2 Jet starts. "DO NOT ENERGIZE JET ENGINES WITH THIS UNIT" shall be stenciled on the control panel in letters ½ inch high. This restriction may be modified in the instruction manual to permit starting jet engines which are energized by 3 phase ac motors.

3.16.2.3 Input rating. A reversible plaque shall be mounted on the control panel near the start button. On one side the plaque shall read 220 volts +10% input and on the other side it shall read 440 volts +10% input. The plaque shall be mounted with machine screws so that it can be removed and reversed. Reversing this plaque shall be part of the standard operating procedure for changing the input voltage connection outlined in the instruction book.

3.17 Workmanship. The unit, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts and tightness of assembly screws and bolts.

3.17.1 Cleaning. The unit shall be thoroughly cleaned. Loose solder, metal chips and other foreign material shall be removed during or after final assembly.

3.18 Instructions. Brief instructions for operating the converter, together with the electrical schematic diagram shall be permanently mounted on or near the control panel. The instructions and diagram shall be protected against elements that will affect their legibility.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

4.2 Classification of tests. The inspection and testing of converter sets shall be classified as follows:

a. Preproduction testing (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 Preproduction testing. The contractor shall subject the first article produced to all tests listed in paragraph 4.3.2, and this article plus an additional three units shall be subjected to the reliability demonstration listed as paragraph 4.3.2o. The additional three units shall be subjected to the quality conformance inspection described in 4.4 prior to start of reliability testing.

4.3.1 Preproduction test report. After the converter has satisfactorily passed all of the preproduction tests, the contractor shall prepare a test report in accordance with MIL-STD-831 and submit it to the procuring activity for approval. The test report shall contain results of the preproduction tests and confirmation that the converter conforms to the requirements of the specification. Copies of all log sheets showing recorded values, adjustments, notes, et cetera and certification that the converter meets the requirements for which no test is given or for which test has been waived shall be included.

4.3.2 Preproduction tests. Preproduction tests shall consist of all the tests specified under 4.6 and shall be conducted in the following order:

a. Examination of product (see 4.6.1).
b. Running-in (see 4.6.2).
c. Individual test run (see 4.6.3).
d. Component balance (see 4.6.4).
e. Dielectric strength (see 4.6.5).
f. Efficiency (see 4.6.6).
g. Voltage adjustment (see 4.6.7).
h. Transportability (see 4.6.8).
i. Mobility (see 4.6.9).
j. Initial calibration (see 4.6.10).
k. Electromagnetic interference (see 4.6.19).
l. Noise (see 4.6.20).
m. Environmental tests (see 4.6.21).
n. Final calibration (see 4.6.22).
o. Reliability demonstration (see 4.6.23).

4.3.2.1 Rejection and retest. In the event of failure of the first article during preproduction testing, the contractor shall stop the testing until failure analysis is perfrmed by the contractor and submitted to the Government procuring contracting officer for evaluation and decision for procedure.

4.4 Quality conformance inspection. Each converter produced shall be subjected to the following tests described under 4.6:

a. Examination of product (see 4.6.1).
b. Running-in (see 4.6.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

c. Individual test run (see 4.6.3).
d. Component balance (see 4.6.4).
e. Dielectric strength (see 4.6.5).

4.4.1 Rejection and retest. When one or more units from a production lot fail to meet the specification, acceptance of all units in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated. The cost of all necessary corrective action is the responsibility of the contractor.

4.4.1.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of an individual test failure. Final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.4.1.2 Defects in converter units already accepted. The investigation of a test failure could indicate that defects may exist in units already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them. The contractor shall correct these defects at no cost to the Government.

4.5 Test conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified, tests shall be made at atmospheric pressure (approximately 29.92 inch mercury) and at room temperature (approximately +25 degrees C). When tests are made with atmospheric pressure or room temperature differing materially from these values, proper allowance shall be made for the difference from the specified condition.

4.5.2 Apparatus. Apparatus used for testing herein shall be of laboratory precision type and shall meet the requirements of MIL-I-45208.

4.6 Test methods.

4.6.1 Examination of product. The converter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, and use of standard parts.

4.6.2 Running-in. The nature and extent of running-in shall be determined by the manufacturer. All necessary adjustments shall be made during this run and shall remain undisturbed thereafter except where specified.

4.6.3 Individual test run. Input voltage shall be 220 volts. The unit shall be started, output voltages set at rated values at no load, and the loads then increased to full load on both generators. The alternator load shall be at 0.8 lagging pf. The unit shall be operated for 24 hours, the first 16 hours at full load, and the last 8 hours at intermittent overload cycle described in 3.6.1.1.1. Hourly readings shall be made of dc and ac output voltages, currents and frequency. In event of unit failure during test run, contractor shall repair unit and repeat test run. Contractor shall keep records of all failures occurring during test runs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

4.6.4 Component balance. The motor-generator shall be started and run at no load. Using a velocity type pickup and vibration meter, measurements of linear vibration shall be made along the horizontal and vertical axis of both the motor and generator. All other components shall be observed and shall exhibit no excessive vibration. The test shall be repeated at full load at 0.8 pf and the vibration shall not exceed the limit specified in 3.5.5.

4.6.5 Dielectric strength. While the unit is hot as a result of operating, the motor, alternator and power transformers shall withstand the following voltage at commercial frequency applied between windings and frames. No breakdown shall result from this test.

a. Windings of 50 volts and less - 500 volts rms for 1 minute or 600 volts rms for 1 second.
b. Windings over 50 volts - twice rated voltage plus 1,000 volts rms for 1 minute or 120% of 1 minute value for 1 second.

4.6.6 Efficiency. The converter shall be operated at 100% loads and 0.8 lagging power factor on rated input voltage. Readings of input and output watts shall be measured to determine the efficiency. Input watts shall also be measured and recorded when the converter is operating with no loads.

4.6.7 Voltage adjustment. With the converter operating at no loads, the voltage adjustments shall be varied through sufficient range to show that the output voltages can be varied from 90 to 110 percent of rated values at 90 and 110 percent input voltages.

4.6.8 Transportability. Design of the converter to meet the air transportability requirements of MIL-A-8421 shall be demonstrated by inspection and analysis, as appropriate.

4.6.9 Mobility. The converter set shall be subjected to mobility tests in accordance with MIL-M-8090 for type I, group B, class 2 mobility except the maximum towing speed shall be 10 mph.

4.6.9.1 Brake. The unit shall be placed facing downward on a 15 degree incline. The towbar shall be raised to the vertical position. Without manual attention thereafter, the brake shall remain set and the unit shall not move down the incline. The test shall be repeated with the unit facing up the incline. The unit shall then be placed on a level surface and it shall be demonstrated that braking action starts when the towbar reaches a height of between 40 and 48 inches above the surface.

4.6.10 Initial calibration. The converter shall be connected to a 220 volt source. The unit shall be started and allowed to operate until the output voltages are stabilized. The output voltages shall be adjusted to rated values at no load and shall not be adjusted during the remainer of the tests unless required by the test. The following tests shall be per-

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

formed and the results recorded:

a. Motor inrush current (see 4.6.11).
b. Voltage regulation (see 4.6.12).
c. Voltage modulation (see 4.6.13).
d. Transient voltage recovery (see 4.6.14).
e. Frequency stability (see 4.6.15).
f. Harmonic output (see 4.6.16).
g. Phase voltage balance (see 4.6.17).
h. Ripple (see 4.6.18).

4.6.11 Motor inrush current. Readings shall be made of the motor inrush current at 440 and 220 volts. The maximum current shall not exceed the values specified in 3.7.3. This shall be recorded by a light beam oscillograph. The voltage recorded shall be corrected to rated value in determining actual current values.

4.6.12 Voltage regulation. Both alternating current and direct current output voltages shall be set at rated values at no load. The loads shall then be increased to full loads on both generators. Readings shall then be taken of both output voltages. The loads shall then be reduced to zero and readings again taken. The entire test shall be repeated at both 90 and 110 percent of rated input voltage.

4.6.13 Voltage modulation. An oscillograph trace shall be taken of the alternating current output voltage at no load, ½ load and full load at 1.0 pf and 0.8 pf. The trace shall be long enough and the amplitude of the trace large enough to provide an indication sufficient for measuring any voltage modulation that may be present.

4.6.14 Transient voltage recovery. Oscillograph traces shall be taken of alternating current and direct current output voltages when the load is suddenly changed from no load to full load and from full load to no load. This shall be repeated for conditions of 90 percent and 110 percent rated input voltages. The test shall be conducted with the alternator load at 0.8 pf.

4.6.15 Frequency stability.

4.6.15.1 Measurements. Frequency stability measurements shall be made during sudden application and removal of the following loads:

a. Full ac load, 0,8 pf.
b. Full ac load, 1.0 pf.

4.6.15.2 Sudden changes. Frequency stability measurements shall also be made while suddenly changing input voltage from rated to both 110 percent and 90 percent rated voltage while the following outputs are being supplied by the alternator:

a. Full ac load, 0,8 pf.
b. Full ac load, 1.0 pf.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

4.6.16 Harmonic content. Harmonic content shall be determined for the loads specified in a and b of 4.6.15.1. Harmonic content shall be determined for line-to-line and line-to-neutral voltages.

4.6.17 Phase voltage balance. Measure alternator phase-to-neutral voltages at no load and at full load, 0.8 pf. Calculate average three phase voltage at no load and at full load. With 10KW, 0.8 pf load on any phase and no load on the other two phases, measure phase-to-neutral voltages and calculate average three phase voltage.

4.6.18 Ripple. The direct current output shall be adjusted to rated voltage at no load and then full load applied. Readings shall be taken at no load and full load as specified in 3.9.2. These values shall not exceed 1 volt peak-to-peak.

4.6.19 Electromagnetic interference tests. The unit shall be tested for conducted and radiated interference in accordance with MIL-STD-461, methods CE03, RE02 and RS03. Measurements shall be in accordance with MIL-STD-462.

4.6.20 Noise levels. It shall be demonstrated that the noise levels produced by the converter do not exceed the sound pressure level values specified in 3.6.6 in any octave band at any of the measurement positions specified in 4.6.20.1. The noise level measurements instrumentation shall be as specified in 4.6.20.2. Ambient background noise levels shall be at least 10 db below the octave band sound pressure levels produced by the converter at the time the noise level measurements are performed.

4.6.20.1 Measurement positions. Noise level measurements shall be made at eight positions spaced 10 feet from the perimeter of the converter and approximately 45 degrees apart. All measurements shall be made with the microphone 5 feet above the floor and oriented toward the converter.

4.6.20.2 Instrumentation. The basic required instrumentation shall consist of a sound level meter in conjunction with an octave band analyzer. This instrumentation shall meet the requirements specified in ANSI Standards S1.4 and S1.11, respectively. An up-to-date calibration and frequency response of the entire measurement system, including microphone, shall be available to permit correction of the measured data. The range of frequency response of the system shall be from 45 to 11,200 Hz and the response stated within +2 db. An acoustical calibration shall be used for an end-to-end system calibration from the microphone to the meter. The sound pressure level (in db ref 0.0002 dynes/cm^2) produced by the calibration, when used in conjunction with the particular microphone in the system, shall be known within $\pm$0.5 db.

4.6.21 Environmental tests. The following environmental tests shall be conducted in accordance with the specified procedures of MIL-STD-810 after the initial calibration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

4.6.21.1 High temperature soak. The converter shall be exposed to but not operated in an ambient temperature of 55 degrees C for 36 hours. It shall then be inspected for distortion, cracking and softening deficiencies. Results of this examination shall be included in the test data.

4.6.21.2 High temperature operation. The converter shall be operated in an ambient temperature of 50 degrees C at full load for 3 hours, and then at overload for 3 minutes. Voltages shall initially be set at no load. Readings shall be taken of output voltage and current and the individual temperature rise of the motor and alternator. Calculations shall be made of the efficiency with rated direct current and alternating current 0.8 pf loads. At the conclusion of this test, an inspection shall be made for any distortion, cracking or softening of parts or deterioration of insulation.

4.6.21.3 Low temperature soak. The test specified in 4.6.21.1 shall be repeated at an ambient temperature of -55 degrees C.

4.6.21.4 Low temperature operation. The test specified in 4.6.21.2 shall be repeated at an ambient temperature of -40 degrees C.

4.6.21.5 Rain. The unit shall be subjected to rain in accordance with method 506, Procedure I, MIL-STD-810. The converter shall be operated at full loads at rated input and output voltages for the last 15 minutes only of each 30 minutes rain. At the conclusion of the test period, the test item shall be inspected for evidence of water inside the control cabinet or in any other locations where it could cause malfunctioning.

4.6.21.6 Fungus resistance. The converter shall be subjected to a fungus resistance test in accordance with MIL-STD-810, Method 508.1. This test may be waived provided the following conditions are met:

a. When AN or MS parts are used the manufacturer certifies that the parts comply with the fungus requirements for the item.
b. When paint, spray, coatings, dips, varnishes, or lacquers used on the unit will not support fungus growth. The manufacturer's description of the materials used shall be included in the test report.
c. Wiring used will not support fungus growth. The manufacturer's description of the wire shall be included in the test report.
d. Wedging, taping, or bracing of coils is accomplished before fungustreatment of the component in which the wedge, brace, or tape is installed.

4.6.21.7 Sand and dust. The converter shall be subjected to the sand and dust test in accordance with MIL-STD-810, Method 510.1. At the conclusion of this test and without being disassembled, the unit may be cleaned of sand and dust by use of a vacuum cleaner.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

4.6.21.8 Humidity. The unit shall be subjected to a humidity test in accordance with the specified procedure of MIL-STD-810, Method 507.1. At the conclusion of the test, the converter shall be operated at full loads at rated input and output voltages for 1 hour.

4.6.21.9 Salt fog. The converter shall demonstrate satisfactory operation when subjected to salt fog in accordance with Method 509, Procedure I, MIL-STD-810. The unit shall be operated under these conditions at full loads (0.8 pf) at rated voltages for not less than 1 hour.

4.6.22 Final calibration. The initial calibration tests specified in 4.6.10 shall be repeated.

4.6.23 Reliability demonstration. The contractor shall document achievement of equipment reliability requirements (paragraph 3.5.2) by conducting a reliability demonstration subsequent to successfully completing all other preproduction testing. If a reject decision is reached at any time during the demonstration the test shall be halted, the contractor, at his own expense, shall accomplish the necessary corrective action to eliminate all identified failure modes, and shall restart the reliability demonstration from the beginning. Detailed requirements for this demonstration are listed in subparagraphs 4.6.23.1 thru 4.6.23.7.

4.6.23.1 Procedures. Four frequency converters shall be used for this demonstration. Test cycles of 16 hours on, 8 hours off shall be used, with only equipment "on" time counting towards the total test time of 2,500 hours. Accept-reject criteria are as follows: 1 failure, accept; 2 failures, reject. All four units shall accumulate approximately the same amount of "on" time during the reliability demonstration (625 hours per unit). During each 16 hour operation period, the converters shall be operated at normal input voltage with 0.8 pf lagging loads as follows:

a. 14 hours at 100% rated 3 phase and dc simultaneous loads.
b. 2 hours of operation subdivided into eight fifteen minute cycles. Each cycle will consist of three minutes at 125% rated 3 phase and dc simultaneous loads and 12 minutes at no loads.

4.6.23.2 Test conditions. All units shall be tested at the contractor's facility in an exposed outdoor location at existing ambient temperatures. No vibration or temperature cycling of the equipment is required.

4.6.23.3 Maintenance. One preventive maintenance action will be accomplished on each equipment after it has completed 400 hours of "on" time to demonstrate the contractor's compliance with requirements of paragraph 3.5.3. No other routine or scheduled maintenance will be allowed during the reliability demonstration.

4.6.23.4 Notification of testing. The contractor shall inform the procuring activity at least 30 days prior to the scheduled reliability demonstration of date testing will begin. Once demonstration has begun, the

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

contractor shall notify the Government by telephone of all equipment failures within 24 hours after their occurrence. When the exact time of failure cannot be determined, failure will be assumed to have occurred one minute after last operational check.

4.6.23.5 Records. The contractor shall monitor output voltages, currents and frequency of each unit during testing. The contractor shall keep a separate record of each equipment's failures during the demonstration. As a minimum, this record shall list the equipment operating time between failures, the clock time required to isolate, repair, reassemble and verify correction of the failure, the symptons leading to discovery of the failure, skill level(s) of technician(s) required to make the repair, cause(s) of failure, corrective action necessary (if any) to eliminate future failures of this type, and total manhours required to complete the repair.

4.6.23.6 Completion of demonstration. Upon successful completion of the reliability demonstration, the contractor shall inspect, clean, and check all units for continued proper operation, and then prepare them for delivery to locations specified by the Government.

4.6.23.7 Reports. Results of this demonstration shall be documented in a final report (CDRL DI-R-7034). This report must be approved by the Government before any production units may be delivered by the contractor.

4.6.24 Safety conformance. The converter will be examined and approved by the procuring activity for compliance with safety requirements prior to acceptance.

4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.2 Packing. Packing shall be Level A, B, C or Air as specified by the procuring agency (see 6.2).

5.2.1 Level A. The converter shall be packed as specified by the procuring activity.

5.2.2 Level B or Air. The motor generator shall be shipped in a mobile condition and shrouded with 4 mil clear plastic film.

5.3 Physical protection. Blocking, bracing and bolting as required shall be in accordance with MIL-STD-1186. The drop tests of MIL-STD-1186 shall be performed only when specifically required by the procuring agency.

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:19Z
Check the source to verify that this is the current version before use.

5.4 Marking. Exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Frequency Converter, Mobile, Type MC-1A.

5.4.1 Additional markings not covered by MIL-STD-129. The following markings shall also be added:

INPUT: 220/440V, 3Phase, 60Hertz
OUTPUT: 30KW, 0.8 PF, 120/208V, 3Phase, 400Hertz; 500 amp, 28V DC.

6. NOTES

6.1 Intended use. The type MC-1A frequency converter covered by this specification is intended for use in ground maintenance, calibration, and support for all fighter-interceptor type aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation, packaging, and packing (see Section 5).

6.3 Reclaimed material. The use of reclaimed materials shall be encouraged to the maximum extent possible within the quality limits required by this document and to fulfill compliance with the Resource Conservation/ Recovery Act of 1976 (Public Law 94-580) dated 21 October 1976.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 99

Review Activity:
DLA-GS

Agent:
Sacramento ALC/MMEDO

(Project 6125-F186)

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:19Z
Check the source to verify that this is the currentversion before use.

APPENDIX

DERATING CRITERIA FOR ELECTRICAL AND ELECTRONIC PARTS

10. SCOPE

10.1 Scope. This appendix provides derating criteria for electrical and electronic parts for design considerations to achieve satisfactory reliability.

20. GENERAL

20.1 Derating is the limiting of electrical, thermal, mechanical, and other stresses on a part to decrease the degradation rate and increase the expected life of the part. Derating increases the margin of safety between the operating stress level and the failure level of the part, providing added protection from system irregularities that are unforeseen during design. Since the safety margins of part specification ratings for both threshold and time dependent stresses are generally unknown, parts should be derated to the maximum extent consistent with good design practices.

20.2 Selection of parts. When derating, the designer must compare specification ratings of parts with the requirements of the application, and select parts for which the application stresses will not exceed values determined by applying the derating factors herein to the specifiecation ratings.

20.3 Derated stress. For power characteristic derating purposes, the maximum allowable percentage for rated stress to which a part may be subjected is determined as follows:

a. Determine operating or junction temperature of the part.
b. In MIL-HDBK-217B, at the determined operating or junction temperature of the part, find the stress ratio at which maximum part failure rate is experienced.
c. Multiply this stress ratio by the appropriate stress derating factor from the following tables. This product is the maximum allowable portion of rated stress to which a part may be subjected in the application to attain satisfactory reliability.

20.3.1 Example. A general purpose diode at a junction temperature of 85 degrees C has a maximum failure rate of .020/10^6hours) at a stress ratio of 0.6 (from MIL-HDBK-217B, page 2.2.4-3). From the following tables, stress derating factor for general purpose diode power is 0.5. Therefore, maximum allowable portion of rated stress for the diode application is 0.6 x 0.5 = 0.3 (diode may be subjected to a maximum of 0.3 of its specified power ratings.

20.4 Permissable operating level. For all other derating purposes (voltage, current, frequency, etc.) stress derating factors in the following tables are applied directly to the specification rating of the part characteristic

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

to determine permissable operating levels. For example, a switching transistor may be subjected to no more that 75% of its rated current in an application to attain satisfactory design reliability.

30. SPECIFIC REQUIREMENTS

30.1 Application stress. A listing of all parts whose greatest application stress level exceeds the maximum allowable application stress, with the rationale for the specific application must be approved by the procuring activity before manufacture of the prototype article is initiated.

30.2 Negative report, as applicable, is also required before manufacture of the prototype article is initiated.

TABLE I

Microcircuits

Type Microcircuit	Critical Stress	Derating Factor (see Note 1)	Remarks
Digital	Output Current	.80	
	Operating Frequency	.75	
Linear	Bias Voltage	.70	
	Input Signal Voltage	.70	
	Output Current	.75	
	Operating Frequency	.75	
Voltage	Input voltage (Rated Maximum)	.80	
	Output current (Rated Maximum)	.75	
	Power dissipation (Rated Maximum)	.60	
All Types	Junction temperature	110 deg C Max.	

See page 35 for notes

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

TABLE 2

Diodes

Type Diode	Critical Stress	Derating Factor (see Note 1)	Remarks
General Purpose	Power	.50	
	PIV	.50	
	Surge Current	.50	
	Fwd Current	.50	
Rectifier	Power	.30	
	PIV	.50	
	Surge Current	.50	
	Fwd Current	.50	
Switching	Power	.30	
	PIV	.50	
	Surge Current	.50	
	Fwd Current	.50	
SCR	Power	.30	
	PIV	.50	
	Surge Current	.50	
	Fwd Current	.50	
Varactor	Power	.50	
	PIV	.75	
	Fwd Current	.75	
Zener	Pow	.50	
	Fwd Current	.50	
	Zener Current	.5 $\left[\frac{I_{z_{max}}+I_{z_{nom}}}{2}\right]$	
Reference	Power	.30	
	Fwd Current	.50	
	Zener Current	.5 $\left[\frac{I_{z_{max}}+I_{z_{nom}}}{2}\right]$	
All Types	Junction Temp	110 Deg C max	

See page 35 for notes

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

TABLE 3

Transistors

Type Transistor	Critical Stress (see Note 2)	Derating Factor (see Note 1)	Remarks
General Purpose	Power	.50	
	Current	.75	
	Voltage	.75	
Power	Power	.30	
	Current	.75	
	Voltage	.75	
Switching	Power	.50	
	Current	.75	
	Voltage	.75	
All Types	Junction Temp	110 Deg C Max	
	Leakage Current	See remarks	Consider leakage current to be at least twice rated value.
All Types	Frequency	See Note 3	

See page 35 for notes

TABLE 4

Resistors

Type Resistor	Critical Stress	Derating Factor (see Note 1)	Remarks
Fixed	See Note 7		
Carbon Comp.	Power	.50	
Insulated Film	Power	.50	
Wirewound Prec.			
1.0% tol.	Power	.50	
0.1% tol.	Power	.25	
Wirewound Power	Power	.50	
Thermistor	Power	.50	Uses in other than zero power applications shall specify minimum wattage for the application.
Adjustable	See Note 7		
Wirewound	Rated current	.70	See note 8
Non Wirewound	Rated current	.70	See note 8
All Fixed & Adjustable	Voltage	.80	Max voltage shall not exceed 80% of the Mil-rating.

See page 35 for notes

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

TABLE 5
Capacitors

Type Capacitor	Critical Stress (see Note 4)	Derating Factor (see Notes 1&6)	Remarks
Fixed			
Ceramic	Voltage	.50	
	Current (AC/Ripple)	.70	
	Surge	.70	
Mica, Dipped	Voltage	.60	
	Current (AC/Ripple)	.70	
	Surge	.70	
Glass	Voltage	.50	
	Current (AC/Ripple)	.70	
	Surge	.70	
Porcelain	Voltage	.50	
	Current (AC/Ripple)	.70	
	Surge	.70	
Paper	Voltage	.50	Good low frequency characteristics.
	Current (AC/Ripple)	.70	
	Surge	.70	
Plastic	Voltage	.50	Mylar good to 1 MHz, Teflon, Polystyrene, and Polycarbonate good to 1 GHz.
	Current (AC/Ripple)	.70	
	Surge	.70	
Tantalum, Foil or Film	Voltage	.50	Limit to 70 Deg C amb. temperature.
	Current (AC/Ripple)	.70	
	Surge	.70	
Tantalum, Solid, Dry	Voltage	.50	Limit to 85 Deg C amb. temperature.
	Current (AC/Ripple)	.70	
	Surge protection	3 ohms/volt	
Tantalum, Solid, Wet	Voltage	see remarks	Use of tantalum wet slug capacitors is prohibited except by prior approval of the procuring agency. (see note 5)
	Current (AC/Ripple)	see remarks	
Adjustable			
Ceramic	Voltage	.50	
	Current (AC/Ripple)	.70	
	Surge	.70	
Glass	Voltage	.50	
	Current (AC/Ripple)	.70	
	Surge	.70	
Air	Voltage	.30	
All Fixed & Adjustable	Leakage	see remarks	For all applications consider leakage current as twice rated max value.

See page 35 for notes

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

TABLE 6

Relays and Switches

Part Type	Critical Stress	Derating Factor (see note 1)	Remarks
Load Type			
Capacitive	Contact current	.75	Peak inrush current shall not exceed derated limit.
Resistive	Contact current	.75	
Inductive	Contact current	.40	
Motor	Contact current	.20	
Filament	Contact current	.10	

see page 35 for notes

TABLE 7

Miscellaneous

Part Type	Critical Stress	Derating Factor (see note 1)	Remarks
Transformer and Chokes	Flux saturation level	.75	Parts shall be used at winding temp at least 30 deg C below rating, max voltage between windings or between windings and shield shall not exceed 60% of rating for steady-state conditions or 90% of rating with transients.
	Insulation Breakdown volt.	.50	
	Current	.70	
Connectors	Current, Pin	see remarks	Current derating for connector pins shall be in accordance with criteria for relays & switches, depending on type of loading. For paraelleled pins, sum of current must not be more than 100% rated of lowest rated pin.
	Insulation Breakdown volt.	.50	

see page 35 for notes

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:19Z
Check the source to verify that this is the current version before use.

NOTES

1/ Rationale for use of parts which exceed the maximum allowable application stress shall be submitted to the procuring activity for approval.

2/ Worst-case combination of DC, AC, and transient voltages shall be no greater than derated limit.

3/ Transistors rated at 500 Milliwatt or less in TO-5 or TO-18 packages using one mil or smaller aluminum lead wires with therocompression bonds shall not be used at repetition rates of less than 30 Hz. At 30 Hz and 25 degrees C or less, derate the power dissipation to .20 and limit the input current to 50 mA. Above 25 degrees C, derate an additional 1.33 mW/degree C. Gold lead devices should be used at repetition rates less than 30 Hz.

4/ Voltage equals instantaneous total of DC, AC, Surge, and transient voltages.

5/ When approved for use, tantalum wet slug capacitors shall be subjected to special screens such as temperature cycling and seal leakage.

6/ Manufacturer's derating factors shall be applied before applying these factors.

7/ High density packaging may require further derating if ambient temperatures are increased.

8/ Rated current is defined as:

$$I_R = \sqrt{\frac{P_{max}}{R_{max}}}$$

and by limiting the current to .70 rated current, power is limited to .50 maximum power.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-M-9397E(USAF) Frequency Converter, Mobile, Type MC-1A

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER MATERIAL PROCURED UNDER A [ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO. DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

C160F1

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:19Z
Check the source to verify that this is the current version before use.

FOLD

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

U.S. MAIL

AFLC CASO/LODS
Federal Center
Battle Creek, MI 49016

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:19Z
Check the source to verify that this is the current version before use.